

$N=10$

y	x
0	23
0	42
1	16
0	21
1	.
0	.
0	.
1	.
0	.

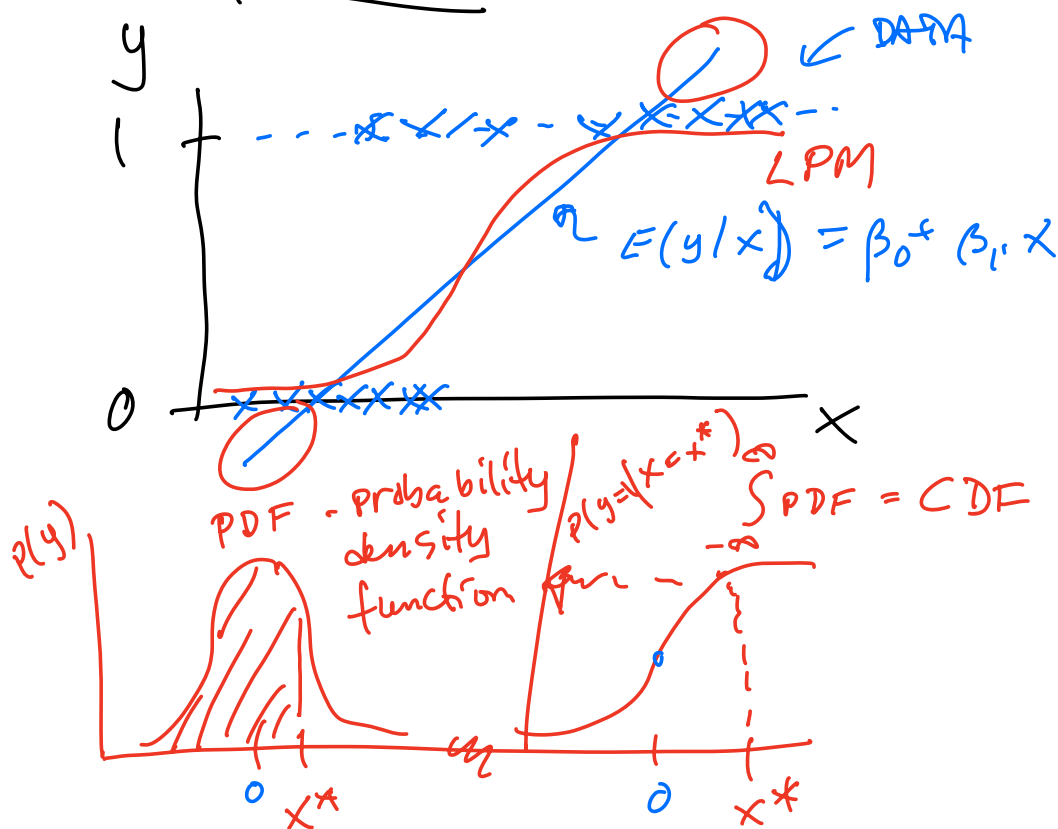
$\{0,1\}$

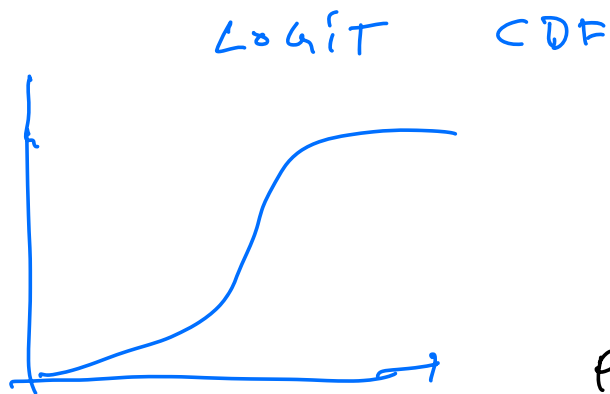
$$P(y=1|x) = \beta_0 + \beta_1 \cdot x$$

$$P(y=1) = \frac{2}{10} = \bar{y}$$

frequencies = probability

$$\text{cor}(x, y) > 0$$





$$P(y=1|x) = z$$

$$z = \beta_0 + \beta_1 \cdot x$$

$$P(y=0|x) = 1 - P(y=1|x)$$

logistische CDF

$$P(y=1|x) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot x)}}$$

↑
 p_i

$$= \frac{e^z}{1 + e^z}$$

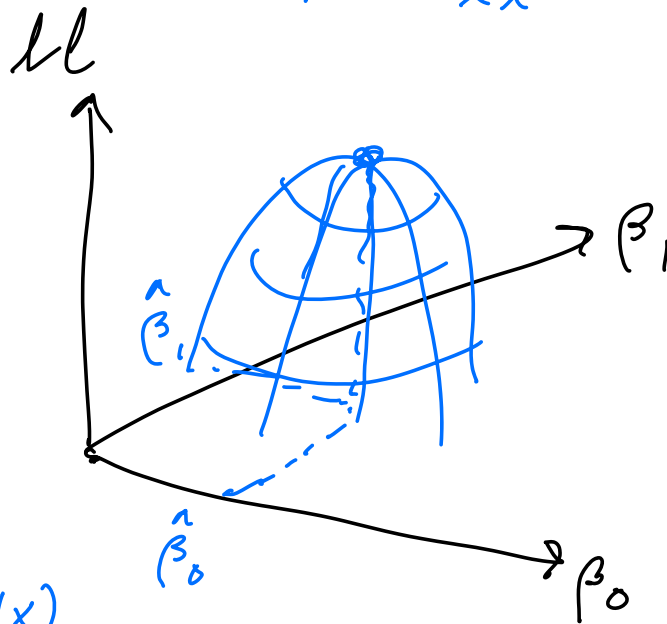
max likelihood for estimate

likelihood (L)

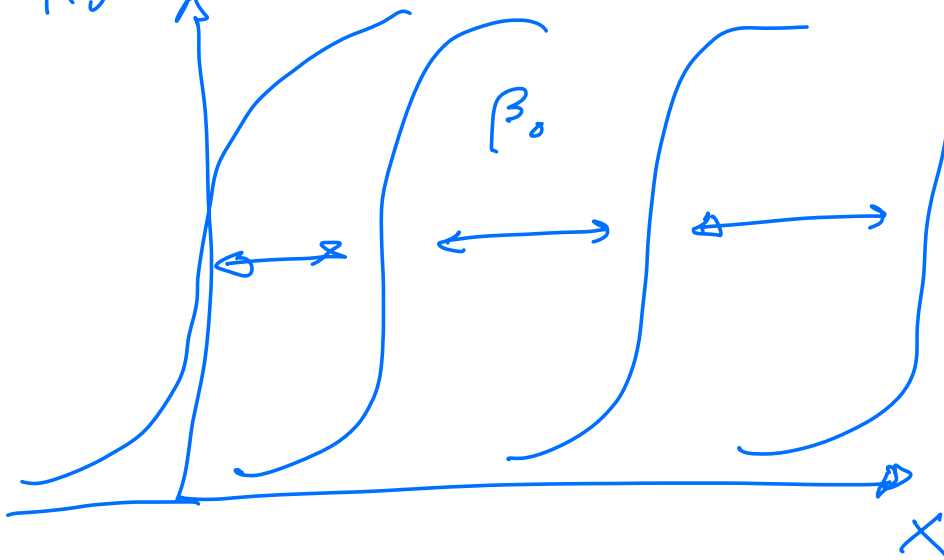
$$l(\beta_0, \beta_1) = \prod_{i=1}^N p_i \cdot \prod_{i=1}^N (1 - p_i)$$

Log likelihood (LL)

$$\max ll \rightarrow \hat{\beta}_i$$



$$P(y=1|x)$$



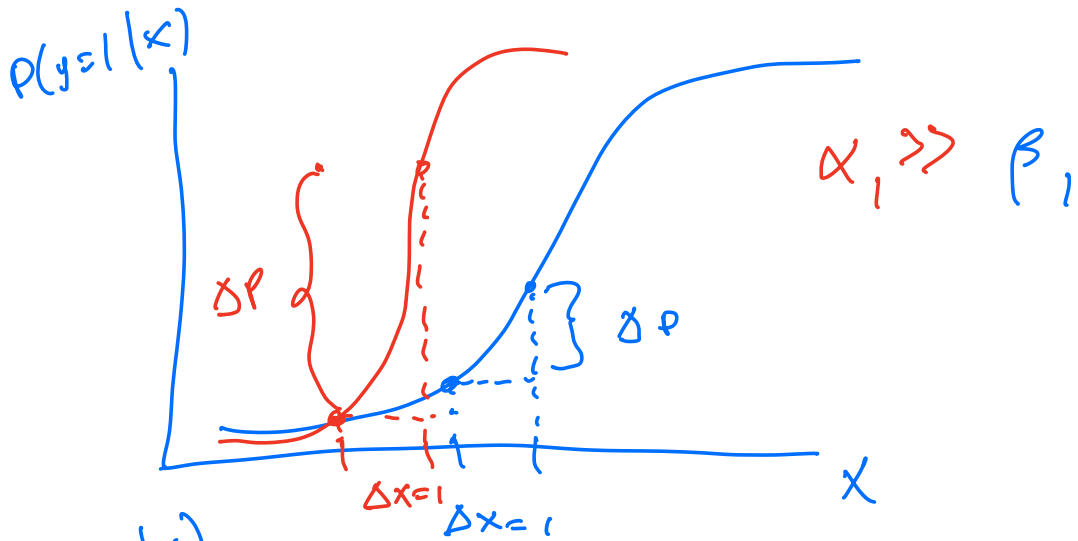
4 → like CDF

$$P(y=1|x) = \frac{1}{1 + e^{-z}}$$

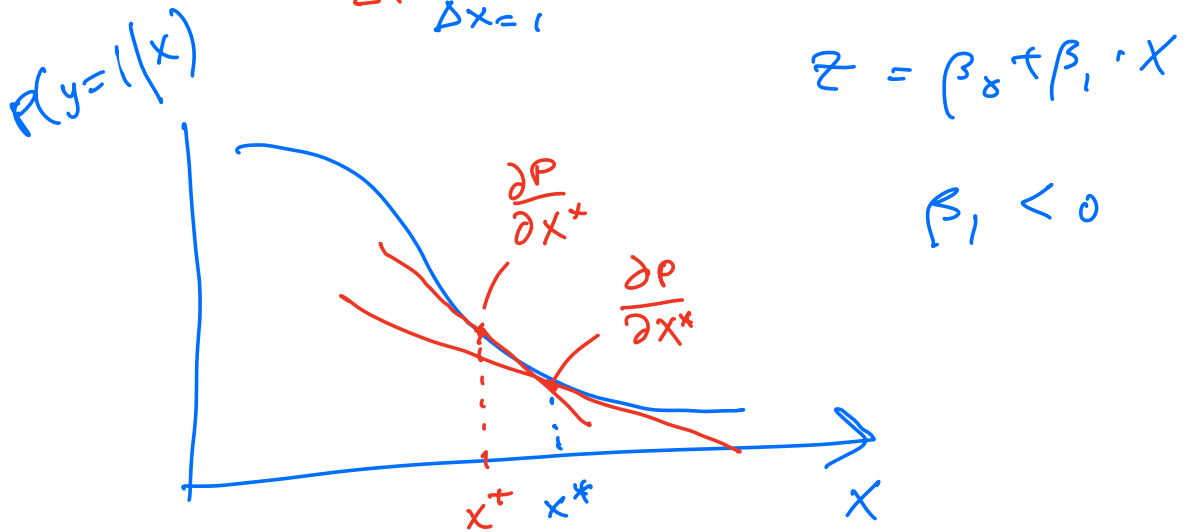
$$z = \beta_0 + \beta_1 x$$

$$z = \beta_0 + \beta_1 x$$

$$z = \alpha_0 + \alpha_1 x$$

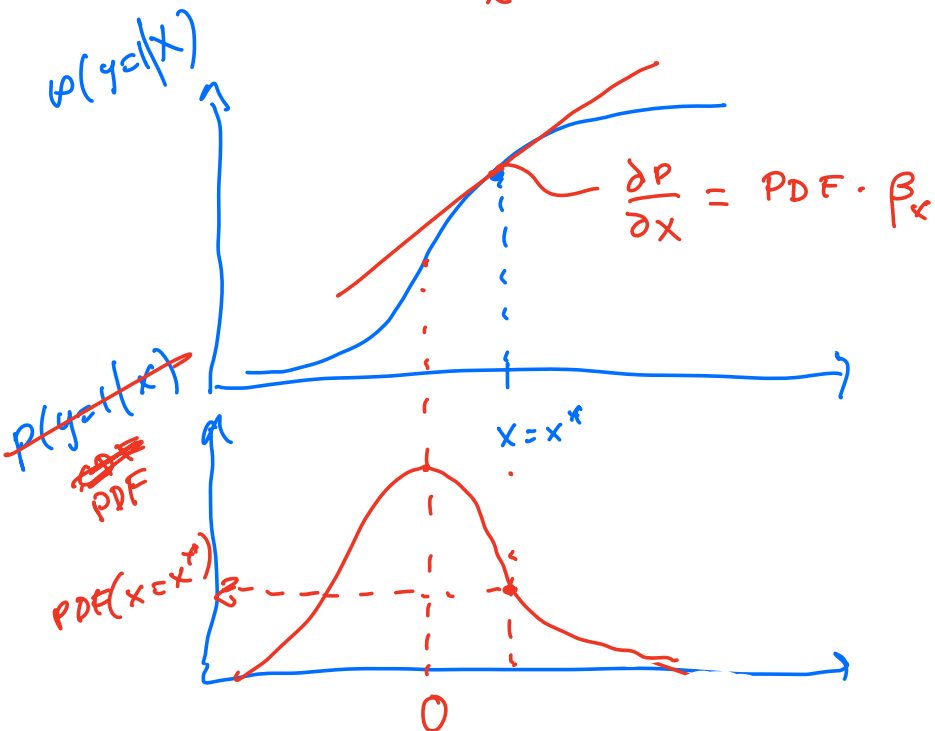


$$\alpha_1 \gg \beta_1$$



$$z = \beta_0 + \beta_1 \cdot x$$

$$\beta_1 < 0$$

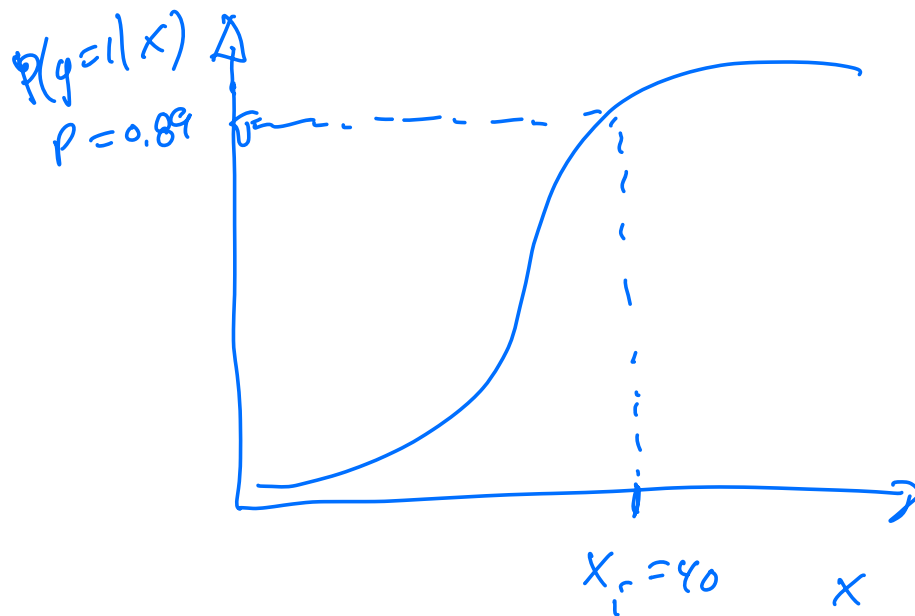


~~$P(y=1|x)$~~
PDF

$\text{PDF}(x=x^*)$

$x=x^*$

0



$$\underbrace{P(y=1|x)}_{\hat{p}} = \frac{1}{1 + e^{-(\hat{\beta}_0 + \hat{\beta}_1 \cdot 40)}} = 0,89$$

Diskret $\{0,1\}$

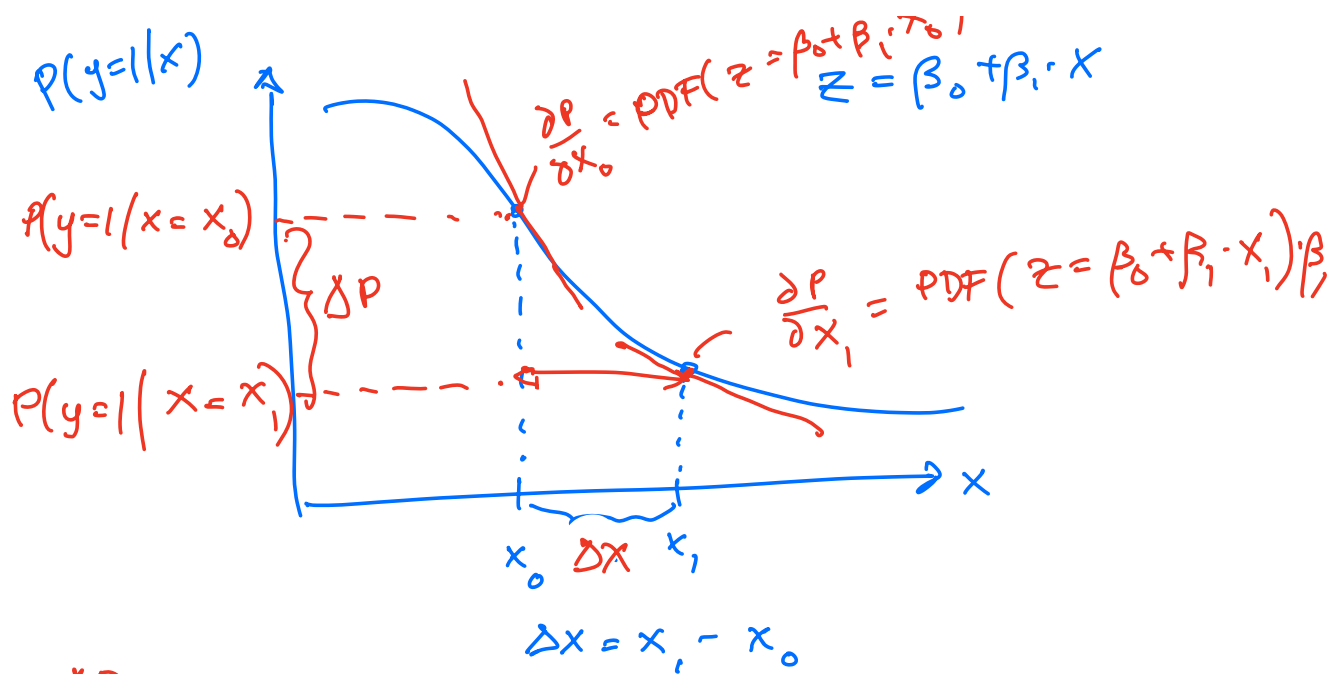
y	$\hat{p}$	x	$\hat{y}$
0	.	.	.
1	0,89	40	1
0	.	.	.

cut-off value $p = 0,5$

$$\hat{p} \geq 0,5 \rightarrow 1$$

$$\hat{p} < 0,5 \rightarrow 0$$

$$\gamma \cdot \beta_1$$



$$\frac{\Delta P}{\Delta x} = \frac{P(y=1|x=x_0) - P(y=1|x=x_1)}{x_0 - x_1}$$

DETTE PRINSIPPET GELDER OGSÅ

FOR x EN EN DUMMYVARIABLE.